LAB 7

Fuzz Testing Using the Hypothesis Library

Objective

To explore fuzz testing as a software security and quality assurance technique by using the Hypothesis library to generate varied inputs, identify failures in the supplied Python functions, and verify fixes through automated tests.

Tools Used

Python 3.13.14, Hypothesis 6.170.15, and pytest 9.2.0.

Functions Tested

Function	Purpose
calculate_dimensional_weight	Calculates package dimensional weight from L×W×H dimensions.
format_tracking_label	Formats a tracking code using an alphabetic prefix and numeric identifier.
parse_package_manifest	Parses JSON package data and extracts the weight value.

Testing Approach and Fixes

The initial Hypothesis tests generated arbitrary text values and None for each function. The tests called the target function inside try/except blocks and failed when an exception was raised. The original implementation produced failures for malformed dimensions, empty or invalid tracking codes, and invalid JSON or manifest inputs.

The application was then updated to validate input types and structure, reject invalid or non-finite numeric values, and return safe fallback values rather than allowing the tested malformed inputs to crash the functions. The fixed functions use 0.0 for invalid dimensional-weight and manifest inputs, and "INVALID" for invalid tracking codes.

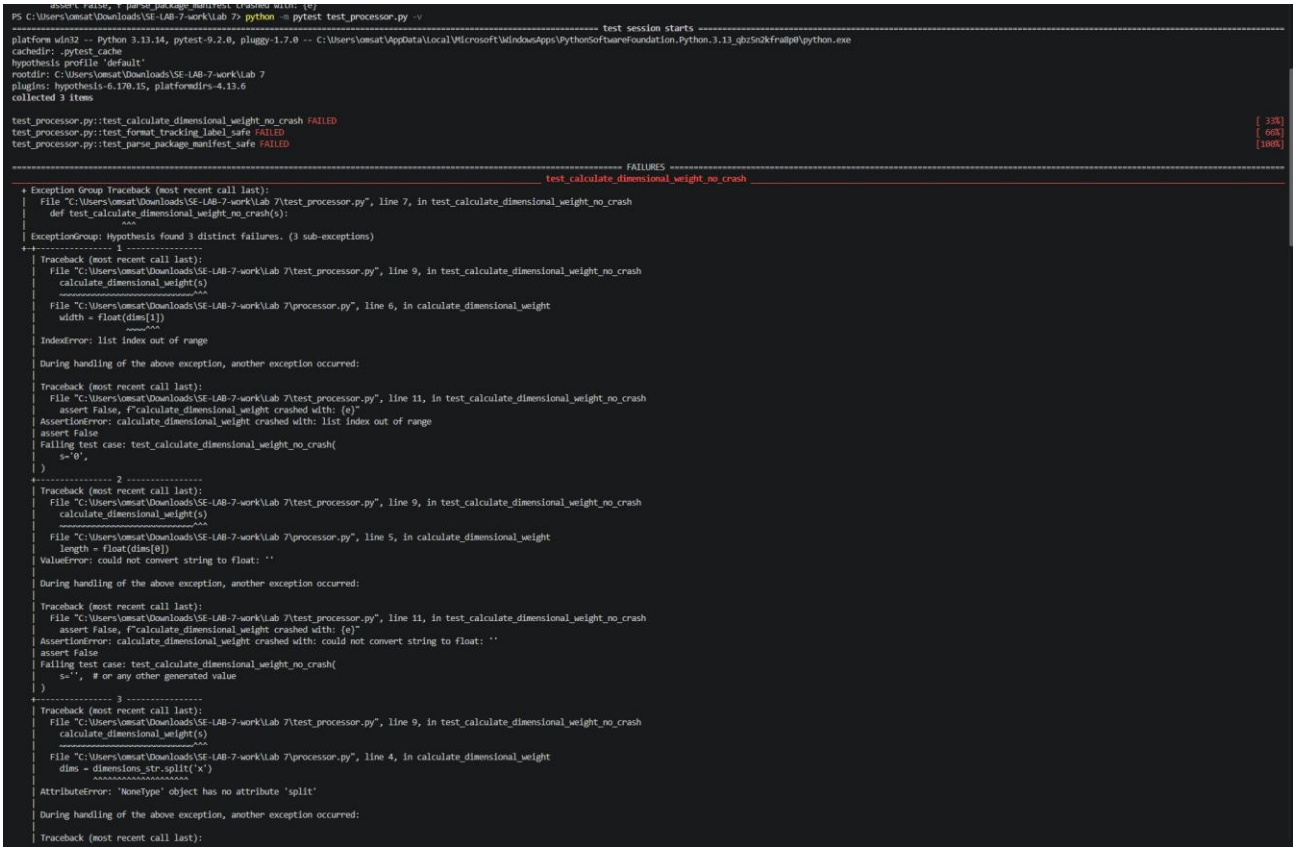
Screenshot 1

```
PS C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7> python processor.py
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): 20x30x10
Dimensional Weight (kg): 1.2

==== Running format_tracking_label ====
Enter raw code (e.g. E123): E123
Formatted Label: E-000123

==== Running parse_package_manifest ====
Enter manifest JSON (e.g. {"weight": 12.5}): {"weight": 12.5}
Manifest Weight: 12.5
PS C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7>
```

Screenshot 2 – Initial Hypothesis fuzz-test failures`



```
PS C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7> python -m pytest test_processor.py -v
============================= test session starts =============================
platform win32 -- Python 3.13.14, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\omsat\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7
plugins: hypothesis-6.170.15, platformdirs-4.13.6
collected 3 items

test_processor.py::test_calculate_dimensional_weight_no_crash FAILED     [ 33%]
test_processor.py::test_format_tracking_label_safe FAILED                [ 66%]
test_processor.py::test_parse_package_manifest_safe FAILED               [100%]

================================== FAILURES ===================================
_____ test_calculate_dimensional_weight_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 7, in test_calculate_dimensional_weight_no_crash
  |     def test_calculate_dimensional_weight_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\processor.py", line 6, in calculate_dimensional_weight
    |     width = float(dims[1])
    |                   ~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 11, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: list index out of range
    | assert False
    | Failing test case: test_calculate_dimensional_weight_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\processor.py", line 5, in calculate_dimensional_weight
    |     length = float(dims[0])
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 11, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_calculate_dimensional_weight_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7\processor.py", line 4, in calculate_dimensional_weight
    |     dims = dimensions_str.split('x')
    |            ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
```

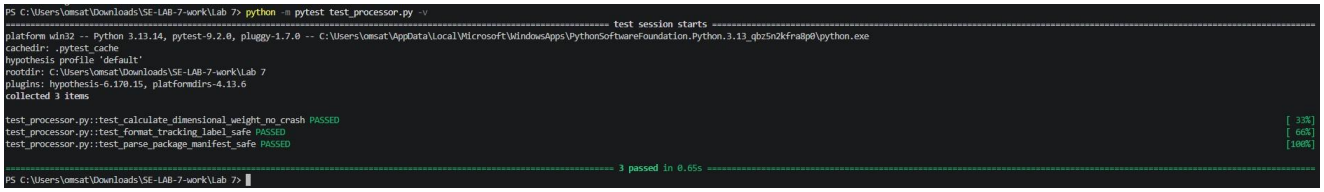
Screenshot 3 – Fixed application execution

```
PS C:\Users\omsat\Downloads\SE-LAB-7-work\Lab 7> python processor.py
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): 20x30x10
Dimensional Weight (kg): 1.2

==== Running format_tracking_label ====
Enter raw code (e.g. E123): E123
Formatted Label: E-000123

==== Running parse_package_manifest ====
Enter manifest JSON (e.g. {"weight": 12.5}): {"weight": 12.5}
Manifest Weight: 12.5
```

Screenshot 4 – Hypothesis tests after fixes



RESULTS AND CONCLUSION

The initial fuzz-testing run exposed failures in all three functions. After input validation and error handling were added, the same test file completed successfully: all three Hypothesis tests passed. This demonstrates how property-based fuzz testing can reveal unexpected-input defects and help verify that defensive changes prevent those crashes.

Final test result recorded in Screenshot 4: 3 passed